

A Perception-Aware Control Framework for Agricultural Quadrotor UAVs: Deep-Learning Crop-Anomaly Segmentation and Treatment Mapping

Ahmed Ashraf Gouda¹, Sergey M. Vlasov²

^{1,2} ITMO University, Saint Petersburg, 197101, Russian Federation

¹ ashraf2024@mail.ru, <https://orcid.org/0009-0005-8548-0449>

² smvlasov@itmo.ru, <https://orcid.org/0000-0002-8345-7553>

Abstract

Introduction. Autonomous agricultural spraying is a coupled perception-and-control problem, yet the control and perception halves are usually studied in isolation. This work extends a five-controller comparative study (PID, LQR, H-infinity, MPC, and SMC) for an agricultural quadrotor UAV with an opt-in perception-and-decision layer and evaluates the controllers jointly with the precision-spraying task they must serve.

Method. A downward-facing RGB camera is simulated during flight; a U-Net with a ResNet-18 encoder, trained on the real Agriculture-Vision CVPR-2020 dataset, labels each captured pixel as background, drydown, weed cluster, or double-plant; affected pixels are projected into a world-frame log-odds treatment map through the camera intrinsics and the UAV pose; and connected affected zones are revisited by a nearest-neighbour planner that logs spray events. The perception output is scored against a real, pixel-aligned ground-truth field mask, so that affected-zone recall, coverage, false-alarm rate, and detection-to-treatment latency measure the controllers on a genuine detection task.

Main results. On the held-out test split the network reaches a mean intersection-over-union (mIoU) of 0.405, reported as the honest real-imagery accuracy rather than the inflated in-distribution score of an earlier synthetic prototype. On the agricultural coverage (lawnmower) trajectory, PID, MPC, and SMC detect every affected zone (recall 1.00) at 88-100 % coverage, whereas LQR and H-infinity leave large gaps and miss half of the affected zones (recall 0.50). On the circular stress trajectory every controller detects all zones, so coverage (31 % for LQR up to 99.5 % for MPC) and false-alarm rate become the discriminating metrics. A 10-seed Monte Carlo campaign with randomized wind, initial conditions, and sensor noise confirms the ranking is statistically stable.

Discussion. High survey coverage does not by itself guarantee treatment success; controller choice propagates all the way to the agricultural outcome. A perception-aware evaluation therefore ranks the controllers on the quantity that actually matters for precision agriculture - how many crop problems are found and treated - and this ranking is driven by each controller's ability to hold its survey path.

Keywords agricultural UAV, precision agriculture, semantic segmentation, U-Net, Agriculture-Vision, treatment mapping, perception-aware control, PID, LQR, H-infinity, MPC, sliding mode control

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Перцептивно-ориентированная система управления сельскохозяйственными квадрокоптерами: сегментация аномалий посевов методами глубокого обучения и картирование обработки

Гуда Ахмед Ашраф¹, Власов Сергей Михайлович²

^{1,2} Университет ИТМО, Санкт-Петербург, 197101, Российская Федерация

¹ ashraf2024@mail.ru, <https://orcid.org/0009-0005-8548-0449>

² smvlasov@itmo.ru, <https://orcid.org/0000-0002-8345-7553>

Аннотация

Введение. Автономное сельскохозяйственное опрыскивание представляет собой связанную задачу восприятия и управления, однако эти две составляющие обычно исследуются отдельно. В работе существующее сравнительное исследование пяти регуляторов (ПИД, ЛКР, H-infinity, MPC и скользящего режима) для сельскохозяйственного квадрокоптера дополнено опциональным слоем восприятия и принятия решений, а регуляторы оцениваются совместно с задачей точечного опрыскивания.

Метод. Во время полёта моделируется направленная вниз RGB-камера; свёрточная сеть U-Net с кодировщиком ResNet-18, обученная на реальном наборе Agriculture-Vision CVPR-2020, классифицирует каждый пиксель как фон, высыхание, скопление сорняков или двойную посадку; поражённые пиксели проецируются в карту обработки в мировой системе координат через параметры камеры и положение аппарата, а связанные поражённые зоны обходятся планировщиком по ближайшему соседу с регистрацией

событий опрыскивания. Результат восприятия сопоставляется с реальной пиксельно-выровненной эталонной маской поля.

Основные результаты. На тестовой выборке сеть достигает среднего значения IoU 0.405. На траектории покрытия ПИД, MPC и скользящий режим обнаруживают все поражённые зоны (полнота 1.00) при покрытии 88-100 %, тогда как ЛКР и H-infinity пропускают половину зон (полнота 0.50). На круговой траектории все регуляторы обнаруживают зоны, и различающими метриками становятся покрытие и частота ложных срабатываний. Кампания Монте-Карло из 10 запусков подтверждает статистическую устойчивость ранжирования.

Заключение. Высокое покрытие обзора само по себе не гарантирует успешной обработки; выбор регулятора влияет на итоговый сельскохозяйственный результат.

Ключевые слова сельскохозяйственный БПЛА, точное земледелие, семантическая сегментация, U-Net, Agriculture-Vision, картирование обработки, перцептивно-ориентированное управление, ПИД, ЛКР, H-infinity, MPC, скользящий режим управления

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Introduction

This paper extends a comparative study of five trajectory-tracking controllers for an agricultural quadrotor UAV - PID, LQR, H-infinity, MPC, and SMC - with a closed-loop perception and decision layer that turns the flight into a useful agricultural mission. In the baseline study the controllers were judged only by how accurately they followed a reference path. Here they are judged by how well the whole vehicle performs the task of finding crop problems from the air and planning treatment for them, because a controller that cannot hold its survey lines fails to image parts of the field, and problems that are never imaged can never be treated.

Trajectory-tracking control of quadrotors has been studied extensively, from foundational modelling and control surveys [1] and geometric tracking on SE(3) [2] to early full-control demonstrations on real platforms [3]; the companion controls study to this work compares the same five control families on agricultural and circular trajectories [4]. The perception layer consists of a simulated downward camera, a convolutional segmentation network trained on real aerial farmland imagery, a projection from image pixels to field coordinates, a probabilistic treatment map, and a revisit-and-spray planner. The contribution is to close the loop in simulation: to run a real segmentation model inside the flight simulation of each controller and to score the resulting treatment decisions against a real ground-truth field, so that the control and perception layers are evaluated jointly rather than in isolation. Three challenges motivate the work. First, segmentation models trained on synthetic or in-distribution imagery report optimistic accuracy that does not transfer to real fields; an honest benchmark must use real imagery and a real ground truth. Second, the perception output must be scored against the same field the vehicle actually flew over. Third, high survey coverage is often assumed to imply mission success, but coverage and treatment precision are different requirements and may rank the controllers differently.

Related Work

The five control families evaluated here rest on classical foundations: optimal linear-quadratic design [5], H-infinity robust control [6], model predictive control [7], and sliding-mode control [8]. Semantic segmentation of imagery advanced rapidly with the U-Net encoder-decoder architecture [9] and deep residual encoders [10], which together form the ResNet-18 U-Net used here; fully convolutional networks [11] and atrous encoder-decoder models [12] established the dense-prediction paradigm these networks belong to. For aerial farmland specifically, the Agriculture-Vision CVPR-2020 dataset provides large-scale, expert-annotated imagery of field anomalies such as drydown, weed clusters, and double-plant regions [13]. Precision-agriculture UAV reviews identify accurate low-altitude tracking under payload and wind variation as the dominant constraint for spraying missions [14, 15], yet they typically treat coverage-path planning, perception, and low-level control as independent modules. The present work couples them in a single closed-loop simulation.

Quadrotor Platform and the Five Controllers

The perception layer is evaluated on top of the same agricultural quadrotor platform and the same five controllers as the companion controls study [4]. The vehicle is a rigid body with six degrees of freedom and four independent inputs, and is therefore an underactuated nonlinear system. Its state is $x = [p, v, q, w]$ in $\mathbb{R}^{13}$, comprising position p , linear velocity v , unit attitude quaternion q , and body angular rate w ; the commanded input is the total thrust f and the body torque τ . Translational motion is actuated only through attitude, so a controller that tracks its survey lines poorly leaves parts of the field unimaged and therefore untreatable - this is the coupling the paper measures.

The dynamics are advanced at $dt = 0.01$ s (100 Hz) by a fourth-order Runge-Kutta step for the velocity and angular rate and a semi-implicit Euler update for the position; the attitude is propagated by the matrix exponential of the midpoint angular velocity and re-projected onto $SO(3)$ by singular-value decomposition. The nominal parameters, held fixed across all five controllers, are mass 1.3-1.5 kg (empty to loaded), arm length 0.225 m, inertia diag(0.0123, 0.0123, 0.0224) kg m², and a per-axis torque limit of 0.5 N m, over a 15 x 10 m field flown at a spray altitude of 4 m and a nominal speed of about 1.2 m/s. Because the plant is identical for every controller, any performance difference is attributable to the control law rather than to the model. The control inputs are hard-limited to the propulsion capability in every control cycle.

The five controllers.

PID uses a cascaded structure: an outer translational loop forms the desired force and an inner attitude loop forms the torque, with anti-windup on the integrators and low-pass filtering on the derivative channels; it is simple and cheap but tuning-sensitive. LQR uses a hover/trajectory linearization and full-state feedback with altitude integral action, obtained by solving the algebraic Riccati equation; it is systematic and smooth but degrades far from hover. H-infinity augments the same linear model with a disturbance input and minimizes the worst-case disturbance-to-performance gain through a bounded-real Riccati iteration plus a disturbance observer; it targets disturbance rejection explicitly. MPC uses a discrete linear model over a prediction horizon of 20 steps and a control horizon of 15, solving a batch quadratic program in closed form and applying only the first move subject to input constraints; it anticipates the reference but is the most computationally demanding. SMC defines translational and rotational sliding surfaces with a boundary-layer saturation that replaces the discontinuous sign to reduce chattering; it is robust to matched disturbances but produces more aggressive actuation. Full gain matrices and synthesis details are given in the companion controls study [4].

Field and ground truth.

The vehicle flies over a 15 x 10 m field whose texture and pixel-aligned ground-truth class mask are both generated from the same real Agriculture-Vision image patches, so that every affected pixel in the mask corresponds to a real anomaly in the imagery the camera sees. The field is built to a realistic affected-area sparsity of 23.1 %; a denser field would saturate recall and hide the differences between controllers. All detection metrics are computed against this real mask.

Evaluation metrics.

Four task-level metrics are used. Affected-zone recall is the fraction of connected ground-truth affected zones that overlap at least one confidently detected treatment cell; it is the primary mission metric, because a missed infestation is more costly than a false alarm. Coverage is the fraction of the field imaged by the camera footprint over the survey. False-alarm rate is the area of detected affected cells that do not correspond to real anomalies, per square metre of field, and must be read jointly with coverage. Detection-to-treatment latency is the mean time between first detecting a zone and logging its spray event. Segmentation accuracy is reported separately as the mean intersection-over-union of the network over the present classes.

Deep-Learning Segmentation Model

The segmentation model is a U-Net with a ResNet-18 encoder [9, 10], trained on the real Agriculture-Vision CVPR-2020 dataset (the combined 2017, 2018, and 2019 miniscale releases) [13]. Four classes are used: background (0) and three affected classes - drydown (1), weed cluster (2), and double-plant (3). A balanced subset of 4500 training, 900 validation, and 900 test patches was assembled; the natural class imbalance of the dataset (background dominates at roughly 82 % of pixels) makes the affected classes intrinsically difficult. The network consumes 256 x 256 RGB tiles and is exported to ONNX for CPU inference inside the MATLAB flight simulation. The model was trained for 25 epochs with the Adam optimizer [16] using a resumable checkpointing scheme; the best validation performance (mean IoU about 0.447) was reached at epoch 24 (Fig. 1).

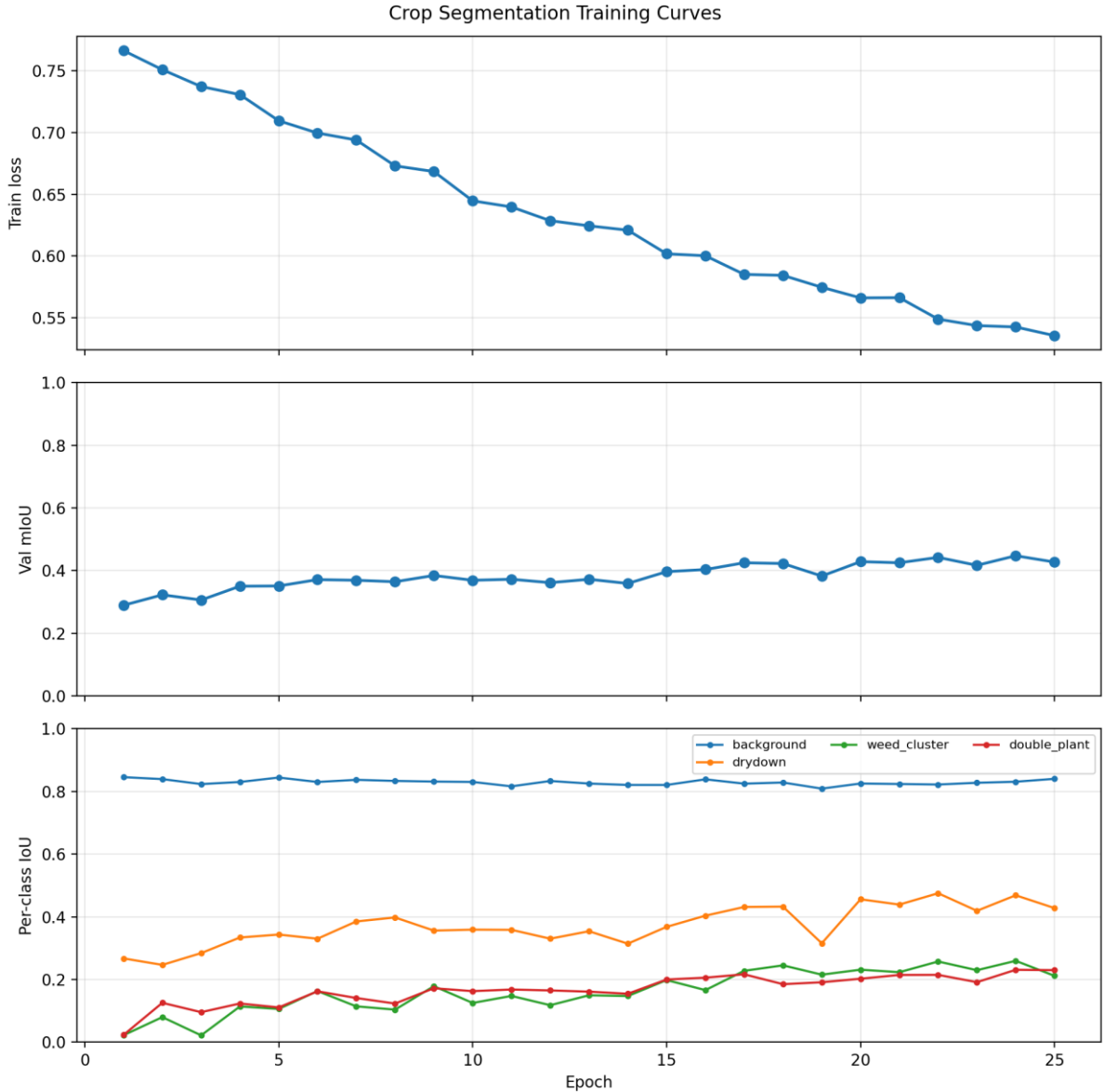


Fig. 1. Training loss, validation mean IoU, and per-class validation IoU over 25 epochs. The affected classes converge more slowly than background because they occupy a much smaller share of the pixels.

Table 1 reports the per-class accuracy on the held-out test split. The model reaches a mean IoU of 0.405. This value is far below the about 0.95 that an earlier synthetic prototype achieved on its own procedural generator; that prototype was trained and tested on the same synthetic process and therefore does not measure field accuracy. It has been removed from the project, and all accuracy claims here are the real Agriculture-Vision numbers.

Table 1. U-Net semantic segmentation accuracy on the held-out Agriculture-Vision test split (mean IoU = 0.405).

Class	IoU	Precision	Recall	Affected
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background	0.812	0.896	0.896	no
drydown	0.422	0.619	0.570	yes
weed cluster	0.206	0.338	0.346	yes
double-plant	0.180	0.300	0.310	yes
mean	0.405	-	-	-

Background is segmented well (IoU 0.812). Among the affected classes drydown is the strongest (IoU 0.422), weed cluster is moderate (IoU 0.206), and double-plant is the weakest (IoU 0.180). The weakness of double-plant is expected: it is the rarest and most visually subtle class in the dataset. Figure 2 shows the per-class breakdown, and Figure 3 shows representative held-out predictions.

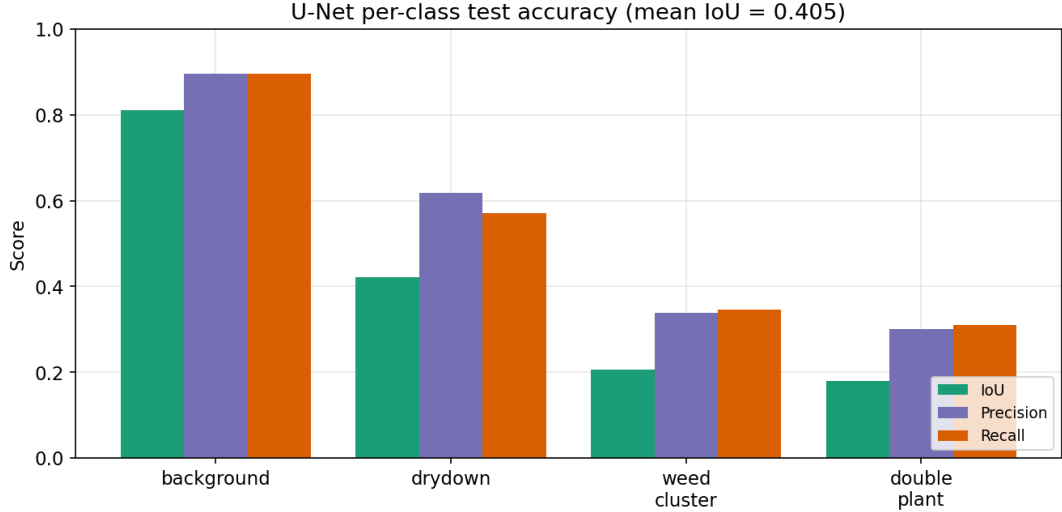


Fig. 2. Per-class intersection-over-union, precision, and recall for the four classes on the held-out test split.



Fig. 3. Qualitative segmentation on held-out Agriculture-Vision tiles. Within each panel: input RGB (left), ground truth (middle), model prediction (right).

Perception-to-Action Pipeline

A downward-facing pinhole camera with intrinsic matrix K captures the field at the spray altitude of 4 m and a frame rate of 5 FPS. Each detected affected pixel (u, v) is back-projected onto the flat field plane $z = 0$: writing the pixel in homogeneous form, its viewing ray in the world frame is $r = R K^{-1} [u, v, 1]$, and the intersection of that ray with the ground plane, with scale $s = -p_z / r_z$, gives the world coordinate. This maps every affected pixel into the 15 x 10 m field. The class-identifier convention is kept symmetric across the Python model (zero-based ids) and the MATLAB pipeline (one-based ids).

Projected detections are accumulated into a field-frame grid of 0.25 m cells using a clamped log-odds accumulator: each confident detection increments a cell's log-odds by +0.85, clamped to $[-10, 10]$. The posterior probability that a cell is affected is recovered by the logistic map, and a cell enters a treatment zone when $P \geq 0.60$. Connected affected cells are grouped into zones by connected-component analysis, and only components of area at least 0.50 m² (eight cells) become treatment zones, which suppresses isolated per-pixel errors. Scoring the treatment map against the real ground-truth mask yields the affected-zone recall and false-alarm rate. A ground-truth zone counts as recalled when any confident detected cell overlaps it, which is robust to adjacent detections merging into a single component.

After the coverage survey, the confirmed zones are ordered by a nearest-neighbour heuristic starting from the vehicle's final survey position, and the revisit reference holds over each

zone long enough for the controller to settle within the spray radius before a spray event is logged. The spray executor is opportunistic: it services every zone the vehicle passes within the spray radius of, so that a single hard-to-reach zone does not block the remaining zones. This design replaced an earlier strictly-sequential executor whose zero settling margin caused controllers that overshoot on the aggressive circular path to leave zones unsprayed; the current executor services every detected zone for all five controllers (Fig. 4).

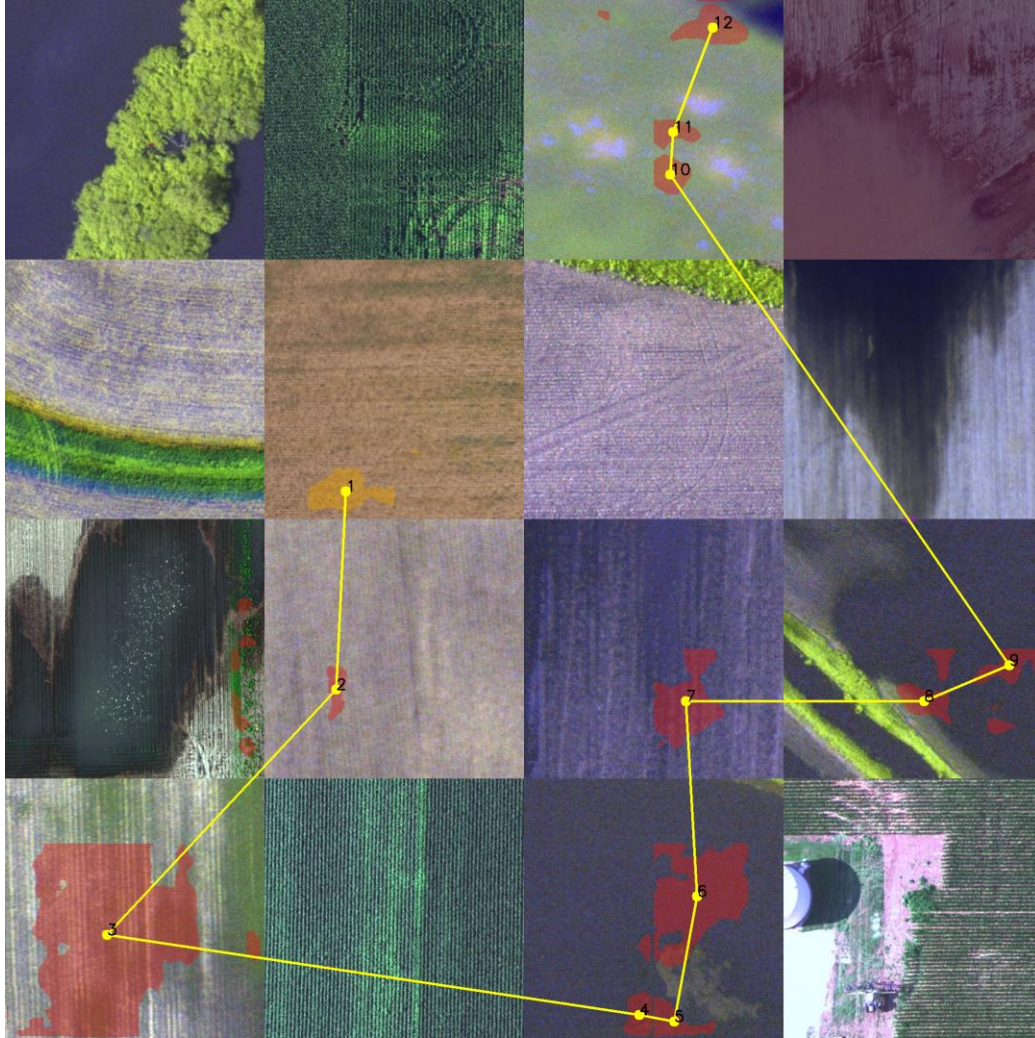


Fig. 4. Nearest-neighbour ordering of the confirmed treatment zones (numbered in service order) overlaid on the stitched real field.

Simulation Results

Real-field detection.

As a first check that the wired pipeline detects real anomalies, a lawnmower survey was flown over the real field and the treatment map scored against the ground-truth mask. The survey imaged 95 % of the field and recalled every affected zone (recall 1.00) at a false-alarm rate of 0.057 per square metre, at an on-board inference cost of roughly 80-150 ms per frame on CPU. Figure 5 compares the real RGB field, the pixel-aligned ground-truth mask, and the model's whole-field prediction. On the whole stitched field the fused segmentation reaches mIoU 0.31, lower than the per-patch test mIoU of 0.405 because the moving survey windows straddle patch boundaries and see partial, off-nadir views; this gap is the expected simulation-to-field effect and is reported openly.

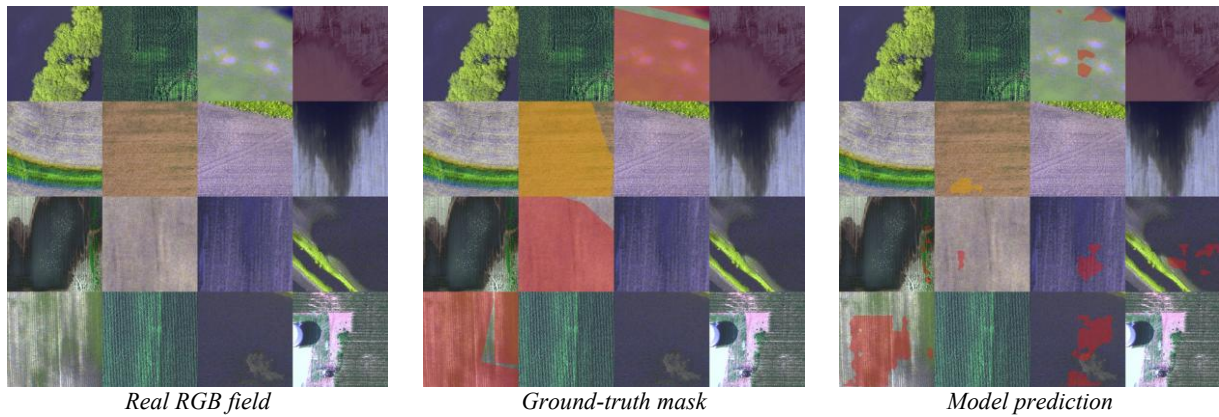


Fig. 5. Whole-field perception over the real 15 x 10 m field. Affected cells are coloured; background is left uncoloured.

Agricultural (coverage) trajectory.

Table 2 reports the five-controller comparison on the agricultural lawnmower survey, scored against the real ground truth, for a single deterministic run. On the coverage mission the controllers separate clearly on the primary metric. PID, MPC, and SMC track the survey lines tightly enough to image 88-100 % of the field and detect every affected zone (recall 1.00). LQR and H-infinity follow the lawnmower pattern poorly, image only 56 % and 62 % of the field, and consequently miss half of the affected zones (recall 0.50). Recall tracks coverage: a controller that cannot follow the survey simply never images some anomalies.

Table 2. Controller performance on the agricultural lawnmower survey (single deterministic run, real ground truth).

Controller	Coverage [%]	Recall	False alarm [1/m ²]	Latency [s]
PID	100.0	1.00	0.137	104.7
LQR	56.4	0.50	0.147	102.5
H-infinity	62.4	0.50	0.105	91.1
MPC	88.2	1.00	0.083	96.6
SMC	100.0	1.00	0.123	101.9

Circular trajectory.

Table 3 reports the same comparison on the circular stress trajectory. Here every controller eventually images enough of the field to detect all affected zones (recall 1.00), so recall no longer discriminates and the discriminating metrics become coverage and false-alarm rate. MPC and PID sweep the most field (99.5 % and 81.0 %) at moderate false-alarm rates, whereas LQR and H-infinity collapse to 32 % and 54 % coverage and carry by far the highest false-alarm rates. With the opportunistic spray executor every controller now services all detected zones, so latency is finite for all five, ranging from 94.9 s (PID) to 145.9 s (LQR); the slower, lower-coverage controllers pay a longer latency because their revisit tours are less efficient.

Table 3. Controller performance on the circular stress trajectory (single deterministic run, real ground truth).

Controller	Coverage [%]	Recall	False alarm [1/m ²]	Latency [s]
PID	81.0	1.00	0.061	94.9
LQR	32.1	1.00	0.280	145.9
H-infinity	54.3	1.00	0.337	125.9
MPC	99.5	1.00	0.158	98.4
SMC	76.6	1.00	0.073	105.9

Controller comparison across trajectories (single deterministic run)

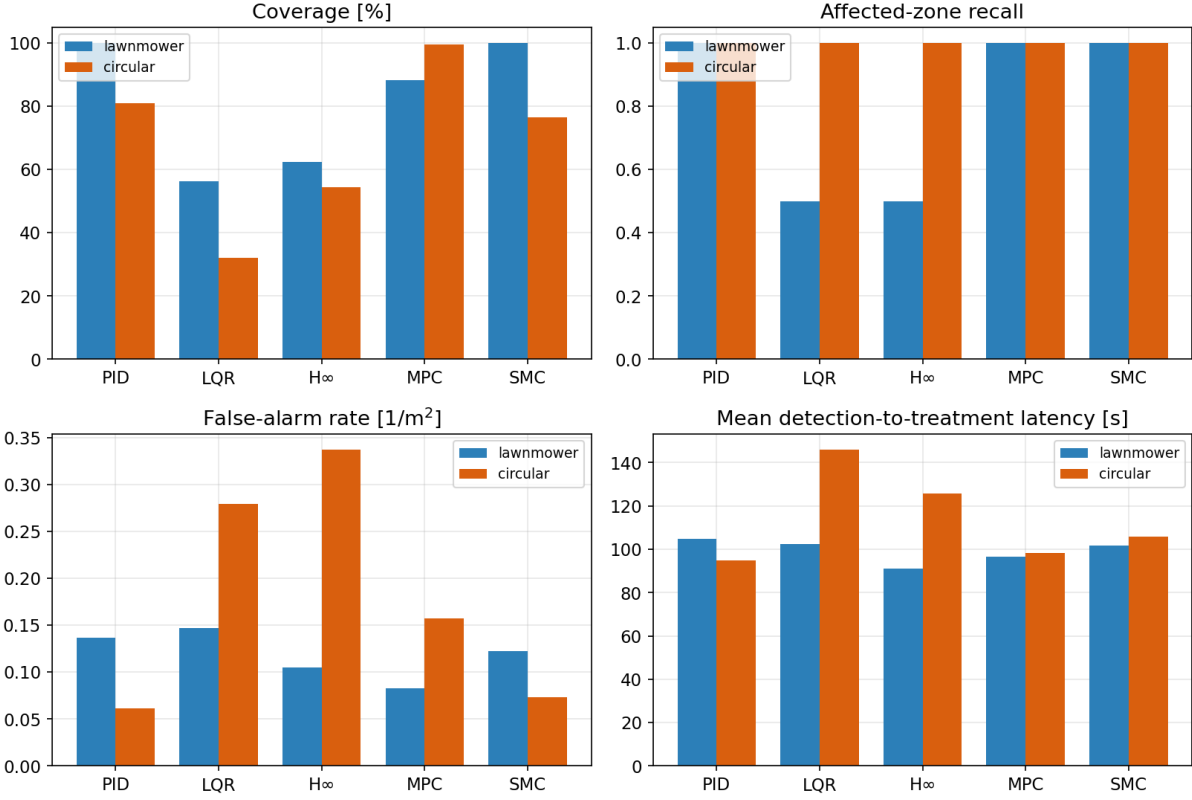


Fig. 6. Coverage, affected-zone recall, false-alarm rate, and mean latency for the five controllers on the lawnmower and circular trajectories.

Monte Carlo campaign.

To confirm that the ranking is not an artifact of a single seed, a 10-seed Monte Carlo campaign was run with randomized wind, initial conditions, and camera noise. Table 4 reports mean coverage and affected-zone recall with 95 % confidence intervals. On the agricultural mission PID and SMC are perfectly reliable (coverage 100 %, recall 1.00, zero interval), MPC is close behind, and LQR and H-infinity are weak. On the circular mission the separation is even starker: LQR and H-infinity not only lose coverage but become unreliable, with recall dropping to 0.20 and 0.75 and wide confidence intervals, whereas PID and SMC hold recall at 1.00 with negligible spread and MPC attains the highest coverage but a more variable recall under stress (Fig. 7).

Table 4. Monte Carlo controller statistics (mean +/- 95 % CI over 10 seeds).

Trajectory	Controller	Coverage [%]	Recall
agricultural	PID	100.0 +/- 0.0	1.00 +/- 0.00
agricultural	LQR	55.5 +/- 2.5	0.55 +/- 0.10
agricultural	H-infinity	62.4 +/- 0.3	0.50 +/- 0.00
agricultural	MPC	86.6 +/- 0.6	1.00 +/- 0.00
agricultural	SMC	100.0 +/- 0.0	1.00 +/- 0.00
circular	PID	81.8 +/- 0.3	1.00 +/- 0.00
circular	LQR	31.2 +/- 6.6	0.20 +/- 0.22
circular	H-infinity	33.9 +/- 11.0	0.75 +/- 0.22
circular	MPC	90.9 +/- 4.3	0.90 +/- 0.13
circular	SMC	70.3 +/- 0.8	1.00 +/- 0.00

Monte Carlo controller statistics (n=10 seeds, 95% CI)

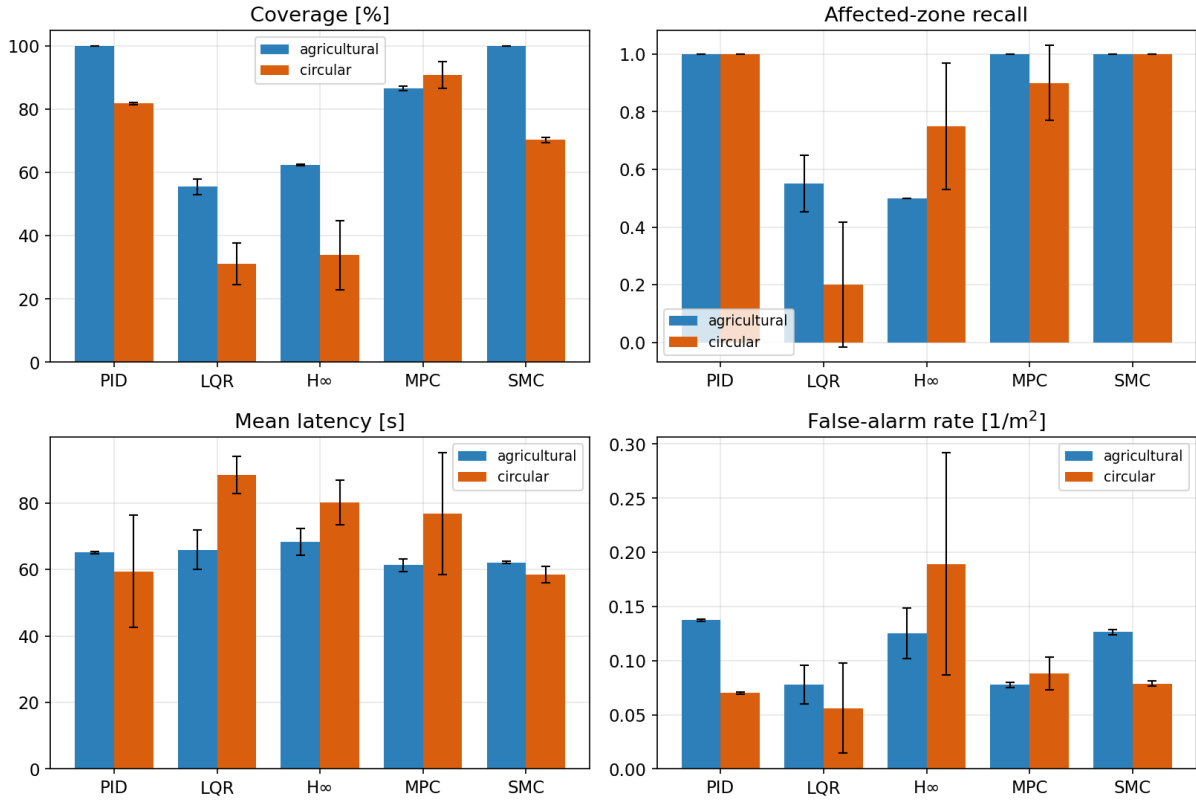


Fig. 7. Coverage and affected-zone recall with 95 % confidence bars for the five controllers on both trajectories.

Perception runtime.

Per-frame ONNX inference runs at roughly 90-150 ms on CPU (Fig. 8), comfortably within the 5 FPS camera budget at survey speed, so perception is not the bottleneck of the pipeline; the limiting factor is the controller's ability to track the survey path, not the cost of the vision model.

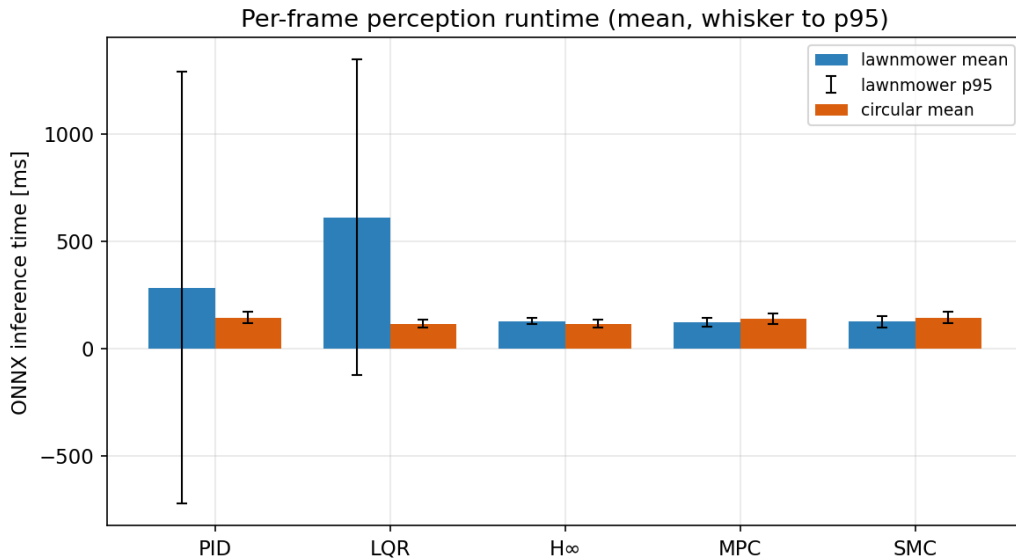


Fig. 8. Mean and 95th-percentile ONNX inference time per frame for the five controllers.

Discussion

The central finding is that survey coverage and treatment success are related but not identical requirements, and that controller choice propagates all the way to the agricultural outcome. On the coverage mission the primary metric, affected-zone recall, separates the

controllers into a strong group (PID, MPC, SMC) that detects every zone and a weak group (LQR, H-infinity) that misses half, purely because the weak controllers cannot hold the survey lines and therefore never image parts of the field. On the circular mission recall saturates and the discriminating metrics move to coverage and false-alarm rate, where the same weak group again trails badly. The Monte Carlo campaign shows this ranking is statistically robust rather than a single-seed accident, and it adds a reliability dimension: LQR and H-infinity are not only worse on average but far more variable under disturbance. A second finding, exposed during integration, is that high coverage does not automatically produce successful spraying: an early strictly-sequential spray executor with no settling margin left zones unsprayed for controllers that overshoot on the aggressive circular path, which is why the executor was redesigned to be opportunistic. Once corrected, the executor treats every zone that any of the five controllers detects, and the reported latency simply reflects how efficiently each controller flies its revisit tour.

Limitations.

Several limitations should be stated openly. The segmentation accuracy is modest in absolute terms (test mIoU 0.405), which is honest for real aerial farmland but far below inflated synthetic scores; the double-plant class in particular is weak and would benefit from more training data or a larger model on a GPU. The whole-field survey metrics are lower than the per-patch test metrics because the moving camera sees partial, off-nadir views; a higher frame rate or overlap-aware fusion would narrow this gap. The camera and field models are simplified - a flat field plane, a real orthomosaic texture, and idealized lighting - so field-deployment claims require on-board validation flights. None of these limitations affect the coverage and recall rankings, which are the principal result.

Conclusion

This work integrated a real-data crop-anomaly segmentation model into the flight simulation of five controllers and evaluated them jointly on a genuine perception-to-action task. Trained on the Agriculture-Vision CVPR-2020 dataset, the U-Net reaches an honest held-out test mIoU of 0.405, and its detections are scored against a real, pixel-aligned field mask. On the agricultural coverage mission PID, MPC, and SMC detect every affected zone at high coverage, while LQR and H-infinity miss half; on the circular stress mission the same weak group collapses in coverage and false-alarm rate; and a 10-seed Monte Carlo campaign confirms that PID and SMC are the most reliable controllers, LQR and H-infinity the weakest and most variable, and MPC the highest-coverage but more variable option under stress. The study shows that a perception-aware evaluation ranks the controllers on the quantity that actually matters for precision agriculture - how many crop problems are found and treated - and that this ranking is driven by each controller's ability to hold its survey path. Future work will strengthen the rare-class segmentation with additional Agriculture-Vision data and validate the pipeline on board a real quadrotor.

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Authors

Ahmed Ashraf Gouda — MSc student, ITMO University, Saint Petersburg, 197101, Russian Federation, <https://orcid.org/0009-0005-8548-0449>, ashraf2024@mail.ru

Sergey M. Vlasov — Associate Professor, ITMO University, Saint Petersburg, 197101, Russian Federation, Scopus ID 55355689600, <https://orcid.org/0000-0002-8345-7553>, smvlasov@itmo.ru